ME-446 · ARTIFICIAL INTELLIGENCE & INTERNET OF THINGS · COMPLEX ENGINEERING PROBLEM · FALL 2026

Detecting dangerous milling conditions from vibration and current

A random-forest safety monitor that flags an aggressive 8 mm radial cut before it damages the tool, spindle or workpiece.

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Department of Mechanical Engineering · Group _____
Dataset: Piecuch & Żabiński, *Scientific Data* 12, 650 (2025) · CC BY 4.0

96.2% dangerous-cycle recall

0.762 MCC at 0.444

968 cycles · 212 dangerous

01 The engineering problem

Radial depth of cut (RDOC) controls how much of the cutter engages the workpiece. At **8 mm** on a 10 mm cutter, chip load and cutting force rise; chatter, overload, accelerated wear and breakage become likely. The operator knows the programmed setting — a process monitor does not.

Task: binary classification of each cycle as optimal (RDOC 4.5 mm) or non-optimal (8 mm) from **120 accelerometer and current statistics**. Metadata, hardness and wear counters are excluded from the inputs.

> **Operational use:** a decision-support alarm. A missed dangerous cut risks the tool and spindle; a false alarm costs an inspection. The threshold is set accordingly.

02 The dataset

968 milling cycles

24 experiments

14 cutting tools

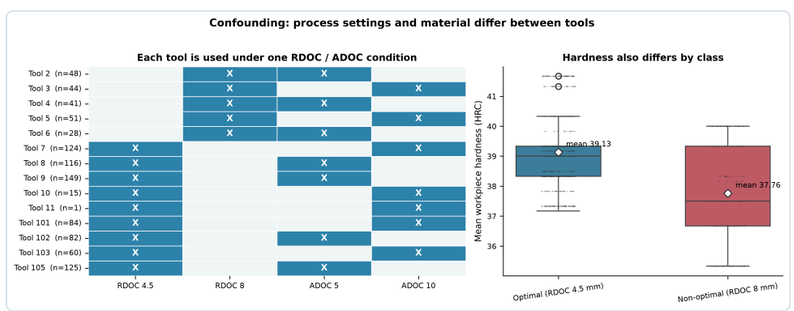
120 sensor statistics

21.9% dangerous class

0 missing values

8 accelerometer channels + 12 current channels × min, max, mean, std, skew, kurtosis. The feature CSV and the metadata workbook match row for row. Fourteen censored cycles and one single-cycle failure are retained, not deleted.

03 The trap: perfect confounding



Every tool and experiment runs at exactly one RDOC/ADOC combination. Tool type 2 never appears at 8 mm.

> **0** experiments and **0** tools contain both classes. A random row split puts the same tool in training and test — it memorises tool identity. Random-split ROC-AUC 0.998; tool-grouped 0.942.

04 Leakage-safe method

1. **Clean and audit.** Parse export quirks, cross-check both sources, keep all 968 cycles.
2. **Features.** signed-log1p on 120 sensor statistics; balanced class weights.
3. **Grouped CV.** Five tool folds; hyperparameters tuned inside each outer fold.
4. **Compare.** Dummy, logistic regression, random forest, histogram gradient boosting.
5. **Select and stress-test.** Threshold on out-of-fold probabilities; leave-one-tool-out and two held-out experiments.

05 Four models, one honest comparison

Model	ROC	PR	Recall	F1	MCC
Dummy (prior)	0.386	0.184	0.000	0.000	0.000
Logistic regression	0.940	0.733	0.948	0.781	0.725
Random forest	**0.942**	**0.688**	**0.892**	**0.799**	**0.742**
Hist gradient boosting	0.936	0.782	0.877	0.727	0.650

5-fold tool-group CV at threshold 0.5. RF wins the operating point; gradient boosting wins threshold-independent PR-AUC — reported, not hidden.

06 Selected model at 0.444

96.2% recall

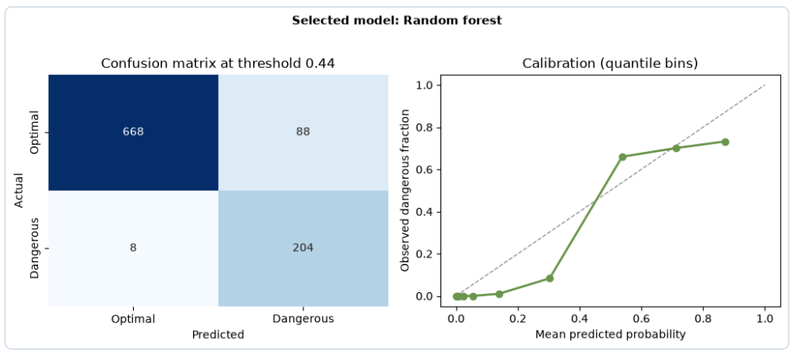
69.9% precision

0.762 MCC

0.810 F1

0.923 balanced accuracy

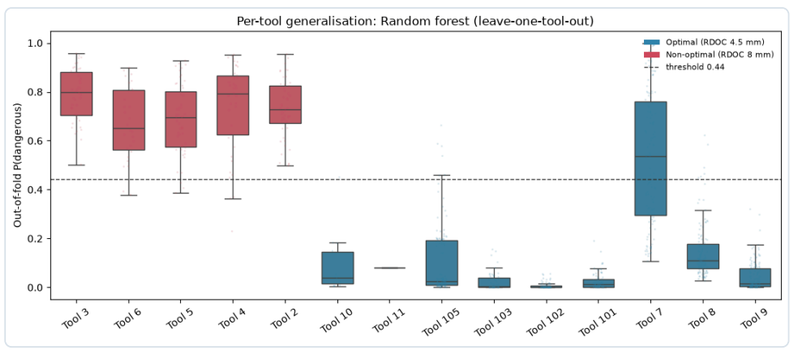
0.082 Brier score



668 true optimal · 88 false alarms · 8 missed dangerous · 204 true dangerous. Probabilities are useful for ranking but over-confident at the top; recalibrate before reading them as exact risk.

Robustness check	5-fold	LOTO
Recall / precision	0.962 / 0.699	0.962 / 0.718
MCC / balanced acc.	0.762 / 0.923	0.778 / 0.928
ROC-AUC / PR-AUC	0.942 / 0.688	0.950 / 0.712

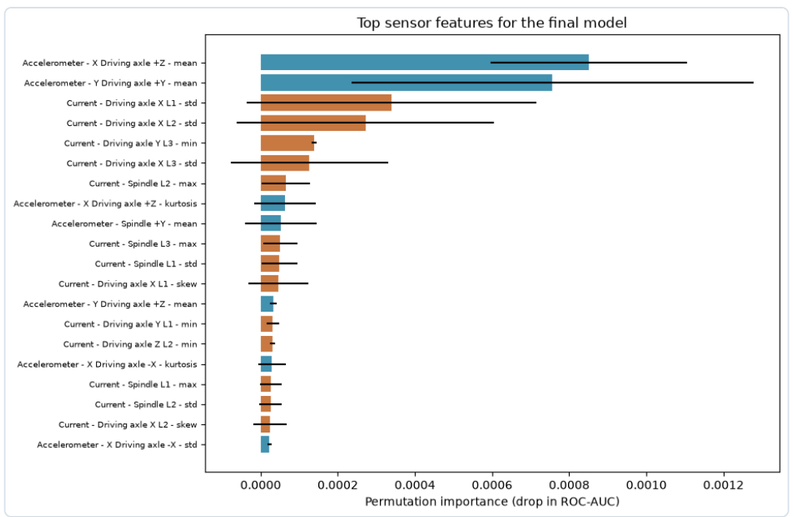
07 Where the errors are



Leave-one-tool-out predictions. Tool 7 (optimal) overlaps the dangerous tools and produces 75 of the 88 false alarms, mostly late in life. The 8 misses are spread over tools 4–6.

> **Residual risk:** an optimal tool can look dangerous when worn. A tool-specific early-life baseline is the most promising fix.

08 It reads force, not the answer key



Permutation importance on the fitted forest. Correlated channels share importance; the ranking is the message.

- **Drive-axis mean acceleration** is the top signal — a static force offset grows with radial engagement.
- **Drive and spindle current variability** follows the heavier, more interrupted cut.
- No identifier, hardness or wear counter is in the feature matrix.

09 Unseen experiments

Held-out	Actual	Cycles	Mean P	Flagged
P002	**Dangerous**	**46**	**0.746**	**97.8%**
P108	Optimal	81	0.006	0.0%

A fresh forest is trained without both experiments. The single P002 miss is the **first cycle of the experiment** (P = 0.439, just below 0.444): a fresh tool has not yet developed the wear-driven signature. Use a rolling window, not a per-cycle verdict.

Final-model scoring: **P002_F12_C4** → 0.967 flag; **P108_F12_C7** → 0.000 clear.

10 Engineering interpretation

A larger radial engagement raises the chip load, cutting force, and therefore drive-axis mean acceleration and current variability. The forest combines these correlated channels into a boundary that catches **96%** of dangerous cycles. The errors are equally physical: wear, hardness and tool geometry push the signature in the same direction as RDOC, which is why one optimal tool overlaps the dangerous class.

> **Consequence:** a false negative continues a dangerous cut; a false positive costs an inspection. Deploy as a human-in-the-loop alarm, not an automatic stop.

11 Caveats and next steps

Limitations

- RDOC confounded with tool, material and wear stage.
- 14 tools, 5 dangerous; tool 7 dominates errors.
- Pre-computed features; raw waveforms unavailable.
- Raw ADC units, one machine, one campaign.
- Threshold is a policy choice.

Recommendations

- Require two consecutive flagged cycles.
- Add a tool-specific early-life baseline.
- Run a balanced experiment at both RDOC levels.
- Calibrate ADC to g and A before transfer.
- Tune the threshold to inspection vs scrap cost.

Data: Piecuch, M. & Żabiński, T. A new open dataset from a milling process — data for classification and estimation of tool life. *Scientific Data* **12**, 650 (2025). DOI 10.1038/s41597-025-04923-y · Data DOI 10.6084/m9.figshare.28589216 · CC BY 4.0.
Method: scikit-learn 1.9 · random forest (150 trees, depth ≤ 6) · nested tool-grouped CV · threshold 0.444.

Code, data and report
github.com/Shahbaz2104/milling-cep
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